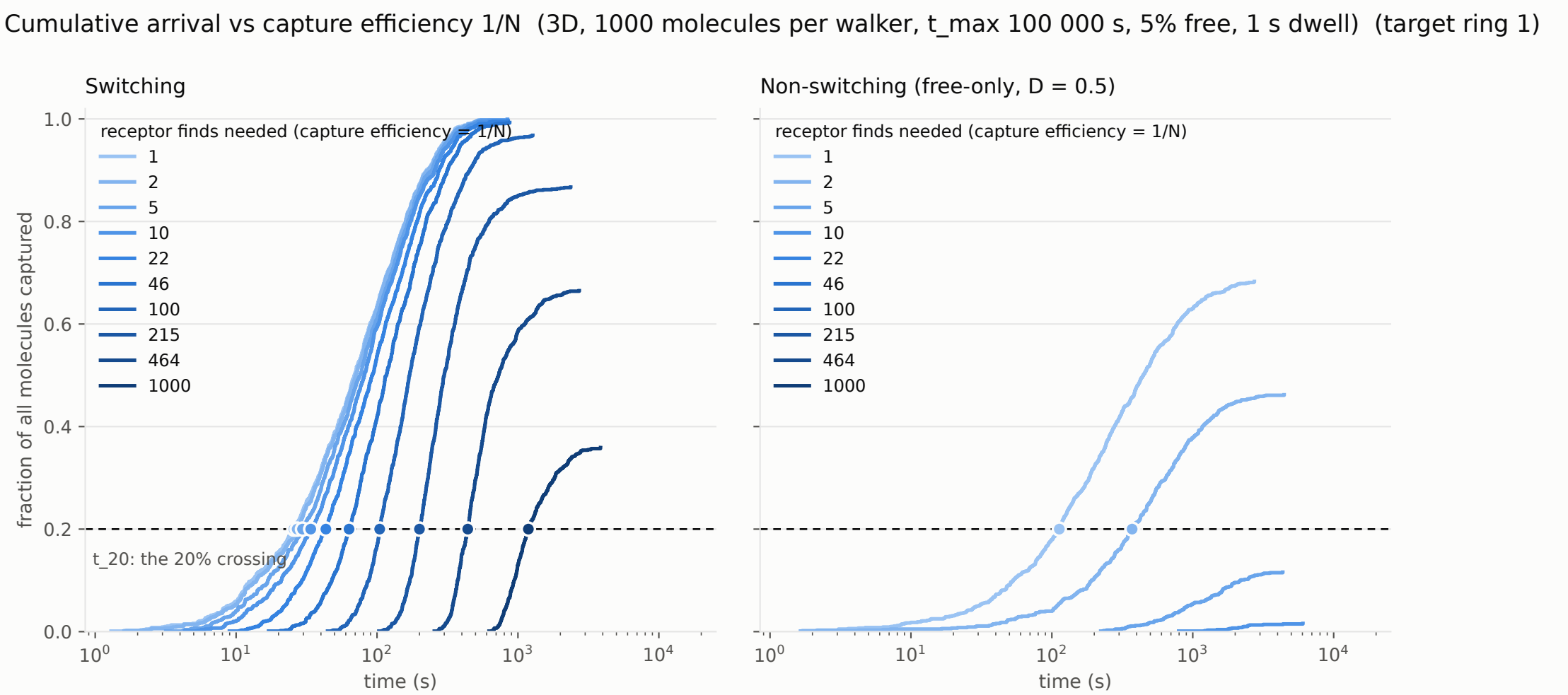

Cumulative arrival vs capture efficiency 1/N (3D, 1000 molecules per walker, t_max 100 000 s, 5% free, 1 s dwell) (target ring 1)
Switching
Non-switching (free-only, D = 0.5)
receptor finds needed (capture efficiency = 1/N)
1
2
5
10
22
46
100
215
464
1000
fraction of all molecules captured
1.0
0.8
0.6
0.4
0.2
0.0
t_20: the 20% crossing
10^0
10^1
10^2
10^3
10^4
time (s)
Each curve levels off at that condition's capture fraction, not at 1: escapes and molecules still searching at t_max never arrive. A curve that plateaus below 0.20 has no t_20.